

Virtualization, Hypervisors & Containers

Why Virtualization Is Security Baseline

You will spend your entire cybersecurity career in virtual machines. They let you:

- Run Kali Linux inside an isolated sandbox
- Snapshot a vulnerable system, attack it, and roll back
- Study malware without touching your host

Hypervisor Types

Type	Example	Description
Type 1 (bare metal)	ESXi, Hyper-V, KVM	Runs directly on hardware, hypervisor is the OS
Type 2 (hosted)	VirtualBox, VMware Workstation	Runs on top of a host OS

Virtual Machine Security Model

The hypervisor creates **guest VMs** – complete computers with virtual hardware. Guests cannot see each other's memory or disks. The **virtualization boundary** is itself a security boundary.

Security insight: Jailbreak/escape exploits ("VM escape") are rare but feared – they break OUT of the hypervisor boundary to hit the host. The famous **VENOM** (CVE-2015-3456) bug lived in the virtual floppy controller of QEMU.

Containers: Shared Kernel, Isolated Userspace

Containers (Docker, containerd) are different – they share the **host kernel** but isolate processes via namespaces and cgroups.

```
VM: [App][OS] | [App][OS] | hypervisor | hardware
Ctr: [App] [App] [App] ----- namespaces/cgroups ----- shared host kernel | hardware
```

Container security:

Namespaces – isolate PID, mount, network, user, IPC

cgroups – limit CPU, memory, devices

Seccomp – restrict system calls

Read-only rootfs – prevent modification

If the shared kernel is exploited, ALL containers on the host are compromised. This is why "container escape" and "run as non-root" are such hot topics.

Your Home Lab Foundation

Every cybersecurity professional's first lab:

Host OS (Windows or Ubuntu) + VirtualBox

Kali Linux VM (attacker)

Windows 10/11 VM (target)

Ubuntu server VM (target + SIEM later)

A target router/firewall VM

Isolation: host-only networking for lab traffic

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